Archives of Public Health

METHODOLOGY Open Access

Sub-national disparities in disease burden of diabetes mellitus and chronic kidney disease in France: a pre-pandemic analysis

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Abstract

Background Diabetes mellitus and chronic kidney disease (CKD) are major global public health issues and the leading causes of mortality and morbidity worldwide. Disability-adjusted life years (DALYs) are one of the most commonly used health gap summary measures in public health and have become an important metric for quantifying burden of disease. The primary objective of this study was to quantify the sub-national burden of diabetes mellitus and CKD in France in 2017 using age-standardized DALY rates (ASDRs), and secondary objective was to assess the relationship between these ASDRs and the French Deprivation Index (FDep), as a measure of socioeconomic disparities.

Methods We used national French health databases to estimate the burden of diabetes mellitus and chronic kidney disease (CKD) in 2017. Disability-adjusted life years (DALYs) were calculated by summing years of life lost (YLL) and years lived with disability (YLD). We calculated ASDRs by applying the age structure of the 2013 European Standard to the observed age-specific rates in France, which were calculated using French population.

Results In 2017, the age-standardized DALY rates per 100 000 population were 685 (95% UI: 552–810) for diabetes mellitus and 251 (95% UI: 205–296) for CKD. ASDRs were lower among females than males (528 vs. 874 for diabetes; 200 vs. 323 for CKD, respectively). The mortality component contributed more to the overall DALYs than the ill-health component, accounting for around 65% for CKD and for diabetes mellitus, the morbidity contributed 55%, in both females and males. Sub-national disparities in DALYs were evident for both diseases, with lower rates observed in metropolitan France compared with the French overseas departments and regions (DROM).

Conclusions These results provide a comprehensive understanding of the disease burden due to diabetes mellitus and CKD in France at sub-national level in 2017, providing valuable insights to inform decision-making processes in health policy. Targeted, evidence-based health interventions and prevention strategies tailored to regional needs can significantly enhance public health outcomes for diabetes mellitus and CKD. Future research should take into account the contribution of various risk factors and determinants of health inequalities to these diseases at regional levels.

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Keywords Years of life lost, Years lived with disability, Disability-adjusted life years, Diabetes mellitus, Chronic kidney disease, FDep, France

Text box 1. Contributions to the literature

- There is limited evidence on the disease burden due to diabetes mellitus and CKD, calculated using local data sources, in France at subnational level.
- This study adds important contribution to the understanding of the disease burden at the regional level in relation to health inequalities before COVID-19 pandemic.
- The contributions of various risk factors and socioeconomic determinants of health to burden estimates at the regional level, using local data sources, need to be assessed.

Introduction

Diabetes mellitus and chronic kidney disease (CKD) are major global public health issues and the leading causes of mortality and morbidity worldwide [1–3]. According to the Global Burden of Disease (GBD) study, the prevalence of these diseases has significantly increased over the past few decades [1, 2, 4, 5]. In 2019, the global number of people living with diabetes was estimated to be 463 million, accounting for 9.3% of the global adult population. By 2030 and 2045, this number is expected to increase to 590 million (10.2%) and 700 million (10.9%), respectively, which will place considerable economic pressure on individuals, families, healthcare systems, and society [5]. CKD had an estimated prevalence of over 697.5 million cases and 1.4 million deaths worldwide in 2019 [6] and is predicted to become the fifth leading cause of death by 2040 [7]. Simultaneously, CKD carries a high risk of accelerated cardiovascular disease and progression to kidney failure with replacement therapy (KFRT), dialysis or transplantation, which is associated with an important burden for patients and healthcare systems, with substantial economic costs [3]. Diabetes mellitus is a leading cause of CKD due to chronic hyperglycemia, which damages renal blood vessels and accelerates nephropathy [8]. In France, nearly half (48%) of all incident dialysis patients have diabetes, although only 5.6% of the general population is treated for it, indicating a disproportionately high risk of renal complications among people with diabetes [9]. The two conditions share common risk factors such as obesity, hypertension, and socioeconomic vulnerability, and their clinical management is closely linked, particularly regarding glycemic control, blood pressure management, and lifestyle interventions [10]. Monitoring these diseases jointly is important for effective prevention, early detection of progression, and integrated care planning [11]. Moreover, the evidence from both cohort study and health claims data consistently showed that diabetes is a significant independent risk factor for CKD in France [12, 13]. These findings strongly support that regional differences in incidence and prevalence of diabetes contribute to regional differences in CKD prevalence.

Health inequalities significantly influence the prevalence and outcomes of both diabetes mellitus and CKD. Recent studies indicate that individuals in lower socioeconomic strata experience higher rates of diabetes mellitus and CKD, often due to limited access to healthcare services, nutritious food, and preventive care. For instance, a 2025 global analysis revealed that the burden of CKD attributable to diabetes is disproportionately higher in low- and middle-income countries, correlating with socioeconomic development levels. Additionally, disparities in healthcare access contribute to delayed diagnoses and suboptimal management of these conditions, exacerbating disease progression. Addressing these inequalities through targeted public health interventions and equitable healthcare policies is crucial to mitigating the impact of diabetes and CKD on vulnerable populations [14].

Burden of disease (BoD) studies are an increasingly popular means to assess the population health impact of different health conditions, comprehensively and comparably according to a standardized concept [15, 16]. The GBD study has been updated and expanded to more diseases, injuries and risk factors, by age, sex and country, since 1996. However, the GBD study has only limited access to data at the national level and necessitating the use of broad/universal assumptions and by applying extensive statistical and modelling techniques. The technical approach of the GBD is complex, both in concept and in the application, with numerous possible methodological alternatives, e.g., life expectancies, disability weights and severity proportions, which have significant influence on resulting estimates of DALYs [17, 18]. Hence, interpretation of the results of burden disease studies requires detailed and in-depth methodological knowledge. Sub-national regional burden of disease analyses are of considerable additional value for the assessment of population health, as they provide ground evidence for guiding and prioritizing health care and prevention measures at local levels. However, the subnational level is crucial for monitoring and implementing targeted actions. To our knowledge, no subnational burden of disease analysis focusing specifically on diabetes mellitus and chronic kidney disease has previously been conducted in France. Therefore, we carried out this analysis to estimate the DALYs of diabetes mellitus and CKD in France at the regional level. This study focused on diabetes mellitus and CKD, due to their high public relevance, their inclusion in the routine French surveillance system, and the availability of reliable data sources. Furthermore, this study may be of interest to an international audience, as it provides insights into regional disparities in chronic disease burden and their association with socioeconomic deprivation in a pre-pandemic context. The primary objective of this study was to quantify the sub-national burden of diabetes mellitus and CKD in France in 2017 using ASDRs and secondary objective was to assess the relationship between these ASDRs and the FDep as a measure of socioeconomic disparities.

Methods

Data sources

We used data from the French National Health Data System (SNDS) to capture diabetes and CKD prevalence data. The SNDS is a large medico-administrative linked data set, which is used for public health surveillance in France [19]. It includes the most updated, individual-level health information about health insurance claims, hospital discharge and mortality records for the entire French population (i.e., 66 million people) [19]. This database has the best representation of the whole French population and it has previously been used to calculate national disease prevalence estimates [20]. We used the French Renal Epidemiology and Information Network (REIN) register to capture the data of individuals with CKD treated by kidney replacement therapy and dialysis [21]. For mortality, we used the statistical database on medical causes of death derived from death certificate collection and coded by the Center for Epidemiology on Medical Causes of deaths (INSERM-CépiDc). Deaths were counted when diabetes mellitus or CKD was recorded as the underlying cause of death. Secondary causes were considered only for the redistribution of ill-defined deaths [22]. The method, which included redistribution of ill-defined deaths to diabetes mellitus and CKD, and the results have been reported elsewhere [22]. The choice of 2017 data was based on its status as the most complete pre-pandemic dataset available for calculating YLLs at the time of the study. To ensure consistency, the same year was used for YLD estimates, allowing a comprehensive assessment of the overall burden of disease as DALYs. Therefore, we used 2017 data as an initial proof-of-concept to develop and validate a methodological framework for generating sub-national DALY estimates from administrative data, which can be applied to more recent datasets in the future.

Study population

We estimated the prevalent cases of diabetes mellitus and CKD of stage I-V, using SNDS database. The case definitions used to extract the prevalent cases for diabetes mellitus and CKD from were based on SNDS and REIN registry. *Diabetes mellitus:* The prevalent cases of diabetes mellitus in 2017, including both Type 1 and Type 2 diabetes, were identified in SNDS using one of the algorithms validated. The cases were defined as pharmacologically treated diabetes, identified by the presence of following ICD-10 code of diabetes: E10 – E14 [23] and/or having "at least 3 different dates of delivery of antidiabetics medications (ATC class A10 excluding benfluorex [A10BX06]) during year n or 2 dates of delivery if at least one large packaging delivered". More detail is provided in the *Supplementary material 1: Additional file 1*). *CKD cases including the stage I – V:* The individual cases of CKD (stage I-V) in 2017 were identified in SNDS database using the RENALGO algorithm, a previously validated and published method [24]. This algorithm was developed by the experts from REDSIAM (Réseau pour mieux utiliser les Données du Système national des données de santé) group [25] and identifies individuals with CKD using combinations of following components in the SNDS: (a) Long-term diseases, (b) Physician claims (consultations), (c) Drug delivery, (d) Biological tests, (e) Diagnosis-related groups, (f) Hospitalization diagnoses, and e) Medical acts [24]. More detail is provided in the *Supplementary material 1: Additional file 1*. The REDSIAM algorithm identifies all CKD patients; however, we excluded those who received a kidney transplant or dialysis, first, to avoid double counting with the REIN registry, and second, because the REIN registry provides more reliable data for these treated cases. Nevertheless, this algorithm does not allow defining the different stages in the absence of results on the level of renal function or urinary abnormality. As mentioned earlier, data on treated cases of kidney transplant and dialysis were obtained from the REIN registry. Each patient who underwent kidney transplant or dialysis was counted once in the registry.

In both data sources (i.e., SNDS and REIN registry), the number of prevalent cases was calculated by five-year age group (with the upper opened-ended group defined as 95 years and above), sex and sub-national regions, for the calendar year 2017. We included all individuals who died and lived with a disability in 2017 and had a recorded residential status in France, covering all 18 administrative regions. These consist of 13 metropolitan regions, located in mainland France and Corsica, and 5 French overseas departments and regions (DROM: French overseas territories): Guadeloupe, Guyane, La Réunion, Martinique and Mayotte. The classification follows the official administrative division used by the French government. We excluded those cases, which lacks information on their residential status (i.e., it amounted to 0.2% for diabetes mellitus and 2.2% for CKD stage I-V, of the total cases recorded in whole French residential population).

Disability weight and severity proportion data

For both diseases, we followed the GBD disease models that include different health states with their relative disability weights according to the GBD 2019 study [26], which are applied globally to all countries *(supplementary material 1: Additional file 2).* For the severity proportions, we obtained the data for diabetes mellitus calculated for France from the GBD 2019 study and for CKD of stage I -V, data were obtained from published literature [27], as our dataset lacked information on CKD staging. The data on severity proportions were applied to the total prevalent cases to distinguish cases by health states and included the following age groups: for diabetes mellitus: <15 years, 15–49 years, 50–69 years, 70 plus; for CKD of stage I –V: < 44 years, 45–74 years and ≥ 75 years *(Supplementary material 1: Additional file 3).* The same severity proportions were applied uniformly across regions. We used the disability weights and severity proportions from the GBD 2019 study, which were the most recent available estimates at time of our analysis.

Statistical analyses

The general approach to estimating the disease burden involves calculating DALYs in a given reference year, are one of the most commonly used health gap summary measures in public health and have become a key metric for quantifying the burden of disease [28]. A health gap measure quantifies the difference between the observed population health state and a hypothetical one where everybody reaches old age free of disease or injury [28]. DALYs summarize the years of life lost due to premature mortality (YLL) and the years of life lived with disability (YLD) into a population-based composite indicator [29]. This metric allows direct comparisons of various diseases and injuries and informs discussions related to the prioritization of preventive and interventional measures, usually based on demographic-specific needs. One unit of DALYs corresponds to one lost year of « healthy life ». DALYs estimates transform standard measures such as prevalence and mortality rates into aggregated or integrated measures, e.g. by applying severity distributions and disability weights (DWs) and thus produce new insights and add value to standard population health assessment.

The mortality-related burden of disease was calculated in terms of standard expected years of life lost due to premature mortality (SEYLL) by multiplying the number of deaths with aspirational life expectancy based on the lowest age-specific mortality rates observed. The GBD standard life table was used, to calculate mortality burden. The methodology is described in detail elsewhere [30]. The morbidity-related burden of disease as *YLD*, which provides a population-based quantification of years lived with health impairments. It was calculated using disease-prevalence estimates, the distribution of severity proportions of a disease in a population and severity specific disability weights [26]. Disability weights are a value between 0 (state of full health) and less than 1 (1 would be equivalent to death). Larger weights correspond to greater levels of disability and result in a higher burden of disease [26].

For DALY estimates, the 95% uncertainty intervals (UI) were calculated by a probabilistic sensitivity analysis where 100 random samples were drawn from uniform distribution of these parameters and simulations were done separately for both diseases, at level 3. The resulting UIs reflect only input uncertainty to ensure the comparability across diseases. The YLD estimates were not adjusted for age-related multimorbidity. The results were aggregated to compute the regional and national mean values with 95% uncertainty bounds. We calculated the age-standardized prevalence rates (ASPRs) and DALY that were reported as absolute values, crude rates and age-standardized DALY rates per 100 000 population for the year 2017 at regional levels. We calculated age-standardised rates by applying the age structure of the 2013 European Standard [31] to the observed age-specific rates in France, which were calculated using French population.

To assess the relationship between social deprivation score of FDep (French Deprivation Index) and ASDRs due to diabetes mellitus and CKD across regions, we performed Spearman's rho correlation test The FDep index is an area-based proxy of social deprivation, and the score is constructed based on the 1 st principal component analysis (PCA) (67–70% of the total variance explained depending on the period) of a population-weighted principal component analysis including four aggregate variables that are the percentage of unemployed individuals aged 15–64 years in the active population; the percentage of workers aged 15–64 years in the active population; the percentage of individuals aged 15 years and over with high-school certificates; and the median income per consumption unit [32]. The values of these four variables were derived from the 2015 census. This score is based on these four variables, which is centered and reduced in order to assess the temporal evolutions. FDep scores are calculated for metropolitan France by Center for Epidemiology on Medical Causes of deaths (INSERM-CépiDc) and not yet developed for the French overseas territories [33]. FDep is used as an indicator of deprivation of participants' place of residence on the municipality level. French overseas territories were excluded from this analysis due to lack of data on FDep score across these areas. First, we calculated the FDep score for each region by aggregating the score for each municipality included in the region and then we performed the Spearman's rho correlation test.

We used STATA software to visualize the results at regional levels as maps with "quintile" approach (i.e., the data were divided into five classes with an equal number of observations) applied by default to define the scales for these maps. This study is reported in adherence with the STROBE statement [34].

Results

We identified approximately three million prevalent cases of treated diabetes mellitus (in 2017, with slightly higher number of cases for males than females (55% vs. 45%). We estimated approximately five million prevalent cases of individuals with CKD, including 98.1% at stages I–V, 1.0% receiving dialysis, and 0.9% with a kidney transplant. We observed more females than males among CKD cases (59% vs. 41%). Moreover, it included approximately 18.5% of females and 23.2% of males with associated diabetes.

The ASPRs were for both diabetes mellitus and CKD were highest among males aged 70 to 74 years, reaching 899 and 993 per 100,000 population, respectively (Table 1 and Table 2). This pattern reflects the increasing burden of chronic conditions with age, particularly among older men.

The burden of diabetes mellitus and CKD in the French population in 2017 was estimated as 456 080 DALY and 173 189 DALY, respectively. There was a higher proportion of male DALY than females for both diseases (56% vs. 44%, for diabetes mellitus; 52% vs. 48% for CKD). The crude rates per 100 000 population translated into a relative 681 DALY for diabetes mellitus, higher among males than for females (785 DALY vs. 583 DALY) and 259 DALY for CKD, slightly higher among males than for females (278 DALY vs. 241 DALY) (Table 1, Table 2). In comparison, the ASDRs per 100 000 population yielded 685 (95% UI: 552–810) for diabetes mellitus with a low rate for females than male (528 [95%UI: 440–626] DALY vs. 874 [95%UI : 708–1046] DALY); and 251 (95%UI: 204–296) DALY for CKD with a low rate for females then male (200 [95%UI: 163–241] DALY vs. 323 [95%UI: 267–387] DALY). These patterns suggest a greater overall burden of these conditions among males compared to females.

The age and sex distribution of disease burden showed similar patterns for both diabetes mellitus and CKD, with DALYs increasing steadily with age and peaking from 70 years and above in both sexes (Fig. 1). This age-related rise reflects the growing impact of chronic conditions in older populations.

The composition of DALYs differed between the two diseases, reflecting their distinct impacts on mortality and morbidity (Fig. 2). For CKD, premature mortality (YLL) accounted for the majority of DALYs (around 65%) in both sexes. In contrast, for diabetes mellitus, morbidity (YLD) made up a greater share of the burden (approximately 55%), highlighting its prolonged impact on quality of life rather than early death (Fig. 2, Table 1 and Table 2).

The regional differences in ASDRs estimates were observed for both diabetes mellitus and CKD (Fig. 3). Metropolitan France showed relatively consistent ASDR distributions, while French overseas territories exhibited notably higher rates for both conditions, highlighting regional health disparities (Fig. 3).

Across metropolitan regions, the ASDRs for diabetes were highest in the North-East (Haute-de-France and Grand-Est: 726–1300), and lowest in the South-West (Occitanie, Bretagne and Pays-de-la Loire: 487–624) (Fig. 3a). In contrast, all French overseas territories showed markedly elevated rates, particularly for Mayotte (i.e., 1920), followed by La-Reunion (i.e., 1468) and Guyane (i.e., 1319), reflecting significant geographic disparities in disease burden.

For CKD, the ASDRs were highest in the North-East (Haute-de-France and Grand-Est: 262–314), and lowest in the South-West (Occitanie, Nouvelle-Aquitaine, Bretagne, and Pays-de-la Loire: 200–232) (Fig. 3b). Among French overseas territories, Mayotte and Guyane showed substantially higher ASDRs (i.e., >657 ASDR), while Martinique, La-Reunion and Guadeloupe had lower burden (i.e., between 277 and 397), though still elevated compared to most metropolitan regions (Fig. 3b). These findings further illustrate the disproportionate burden of CKD in certain geographic areas, particularly in French overseas territoties.

We analyzed the relationship between FDep score and ASDRs across metropolitan regions and found a non_significant positive correlation for both diabetes mellitus (Spearman's rho = 0.401, $p = 0.1744$) and CKD (Spearman's rho = 0.181, $p = 0.5533$) (Fig. 4). This suggests that regional variations in social deprivation were not consistently correlated with ASDRs. For both diseases, higher ASDRs were observed in Hauts-de-France and Grand Est, while lower rates were seen in Bretagne (diabetes mellitus) and Pays de la Loire (CKD). A non-linear relationship, possibly U-shaped, may exist, and regional disparities could obscure direct associations in this ecological analysis (Fig. 4).

Table 1 Prevalence and DALY estimates of diabetes mellitus

Sex		Prevalence			DALY			Age-standardized ratio	
		Count	Age-specific rate	Age-standardized rate	Count	Age-specific rate	Age-standardized rate	YLD: DALY ratio	YLL: DALY ratio
Both sexes	All ages	3,364,763	5026	5152	456,080	681	685	55	45
	< 25	39,492	982	56	3075	77	4	70	30
	25–29	18,407	469	28	1697	43	3	67	33
	30–34	26,779	655	43	2463	60	4	67	33
	35–39	41,226	984	69	3481	83	6	73	27
	40–44	68,328	1577	110	6623	153	11	64	36
	45–49	124,018	2745	192	14,306	317	22	54	46
	50–54	213,669	4737	332	24,549	544	38	63	37
	55–59	325,432	7591	493	37,208	868	56	64	36
	60–64	448,123	11,012	661	52,904	1300	78	62	38
	65–69	557,799	14,103	776	65,836	1665	92	62	38
	70–74	501,667	17,983	899	61,274	2196	110	63	37
	75–79	380,382	17,763	711	52,405	2447	98	54	46
	80–84	325,278	17,321	433	55,476	2954	74	44	56
	85–89	204,549	15,768	237	44,011	3393	51	35	65
	90–94	73,956	11,924	95	23,519	3792	30	24	76
	95+	15,658	9277	19	7252	4296	9	16	84
Females	All ages	1,507,750	4366	4200	201,487	583	528	58	42
	< 25	19,247	976	55	1281	65	4	83	17
	25–29	9761	491	29	688	35	2	88	12
	30–34	13,919	665	43	1192	57	4	72	28
	35–39	20,126	945	66	1346	63	4	93	7
	40–44	30,288	1384	97	2476	113	8	76	24
	45–49	52,799	2311	162	5275	231	16	62	38
	50–54	89,316	3892	272	8954	390	27	73	27
	55–59	135,099	6117	398	13,938	631	41	71	29
	60–64	184,486	8677	521	19,529	918	55	69	31
	65–69	225,971	10,853	597	24,314	1168	64	68	32
	70–74	210,424	14,069	703	24,539	1641	82	65	35
	75–79	173,808	14,486	579	23,236	1937	77	56	44
	80–84	166,302	14,716	368	27,811	2461	62	45	55
	85–89	116,844	13,759	206	25,516	3005	45	34	66
	90–94	47,918	10,712	86	15,717	3514	28	23	77
	95+	11,442	8619	17	5676	4276	9	15	85
Males	All ages	1,857,013	5728	6292	254,593	785	874	53	47
	< 25	20,245	987	56	1794	89	5	61	39
	25–29	8646	447	27	1009	52	3	53	47
	30–34	12,860	644	42	1272	64	4	63	37
	35–39	21,100	1025	72	2135	104	7	61	39
	40–44	38,040	1773	124	4147	193	14	57	43
	45–49	71,219	3189	223	9031	404	28	49	51
	50–54	124,353	5612	393	15,595	704	49	58	42
	55–59	190,333	9158	595	23,270	1120	73	60	40
	60–64	263,637	13,568	814	33,375	1718	103	58	42
	65–69	331,828	17,714	974	41,523	2217	122	58	42
	70–74	291,243	22,508	1125	36,735	2839	142	61	39
	75–79	206,574	21,938	878	29,169	3098	124	53	47
	80–84	158,976	21,258	531	27,665	3699	92	43	57
	85–89	87,705	19,575	294	18,495	4128	62	35	65
	90–94	26,038	15,060	120	7802	4513	36	25	75
	95+	4216	11,700	23	1576	4373	9	20	80

Table 2 Prevalence and DALY estimates of chronic kidney disease

		Prevalence			DALY			Age-standardized ratio	
Sex		**Count**	**Age-specific rate**	**Age-standardized rate**	**Count**	**Age-specific rate**	**Age-standardized rate**	**YLD: DALY ratio**	**YLL: DALY ratio**
Both sexes	All ages	4,525,911	6760	6818	173,189	259	251	35	65
	< 25	126,479	3185	179	879	22	1	36	64
	25–29	73,111	1863	112	579	15	1	53	47
	30–34	97,878	2393	156	942	23	1	52	48
	35–39	109,561	2616	183	1165	28	2	58	42
	40–44	126,187	2912	204	1410	33	2	61	39
	45–49	176,876	3915	274	3355	74	5	55	45
	50–54	242,517	5377	376	4537	101	7	56	44
	55–59	316,587	7385	480	7206	168	11	46	54
	60–64	417,456	10,259	616	9155	225	13	48	52
	65–69	551,211	13,936	766	18,471	467	26	46	54
	70–74	554,087	19,862	993	19,642	704	35	45	55
	75–79	494,957	23,113	925	21,121	986	39	38	62
	80–84	516,470	27,502	688	28,391	1512	38	30	70
	85–89	428,415	33,025	495	30,338	2339	35	22	78
	90–94	224,330	36,170	289	19,342	3119	25	15	85
	95+	69,789	41,347	83	6657	3944	8	11	89
Females	All ages	2,453,647	7105	6634	83,197	241	200	37	64
	< 25	72,672	3744	212	555	29	2	26	74
	25–29	51,570	2592	156	282	14	1	48	52
	30–34	68,459	3269	213	403	19	1	56	44
	35–39	70,658	3316	232	437	21	1	64	36
	40–44	73,955	3380	237	572	26	2	61	39
	45–49	98,622	4317	302	1352	59	4	62	38
	50–54	127,378	5550	389	1875	82	6	61	39
	55–59	156,815	7100	462	2901	131	9	49	51
	60–64	198,478	9335	560	3569	168	10	52	48
	65–69	257,911	12,387	681	6875	330	18	54	46
	70–74	266,242	17,801	890	8379	560	28	45	55
	75–79	253,565	21,133	845	9038	753	30	40	60
	80–84	289,405	25,609	640	13,433	1189	30	31	69
	85–89	263,270	31,002	465	16,558	1950	29	21	79
	90–94	152,384	34,066	273	12,154	2717	22	14	86
	95+	52,263	39,368	79	4814	3626	7	11	89
Males	All ages	2,072,264	6392	7138	89,992	278	323	34	66
	< 25	53,807	2641	146	323	16	1	51	49
	25–29	21,541	1113	67	298	15	1	57	43
	30–34	29,419	1474	96	539	27	2	49	51
	35–39	38,903	1890	132	728	35	2	55	45
	40–44	52,232	2434	170	838	39	3	61	39
	45–49	78,254	3504	245	2003	90	6	50	50
	50–54	115,139	5197	364	2662	120	8	53	47
	55–59	159,772	7688	500	4305	207	13	44	56
	60–64	218,978	11,270	676	5586	288	17	46	54
	65–69	293,300	15,658	861	11,596	619	34	42	58
	70–74	287,845	22,245	1112	11,263	870	44	44	56
	75–79	241,392	25,636	1025	12,083	1283	51	36	64
	80–84	227,065	30,363	759	14,958	2000	50	29	71
	85–89	165,145	36,859	553	13,780	3076	46	22	78
	90–94	71,946	41,613	333	7187	4157	33	16	84
	95+	17,526	48,636	97	1843	5115	10	12	88

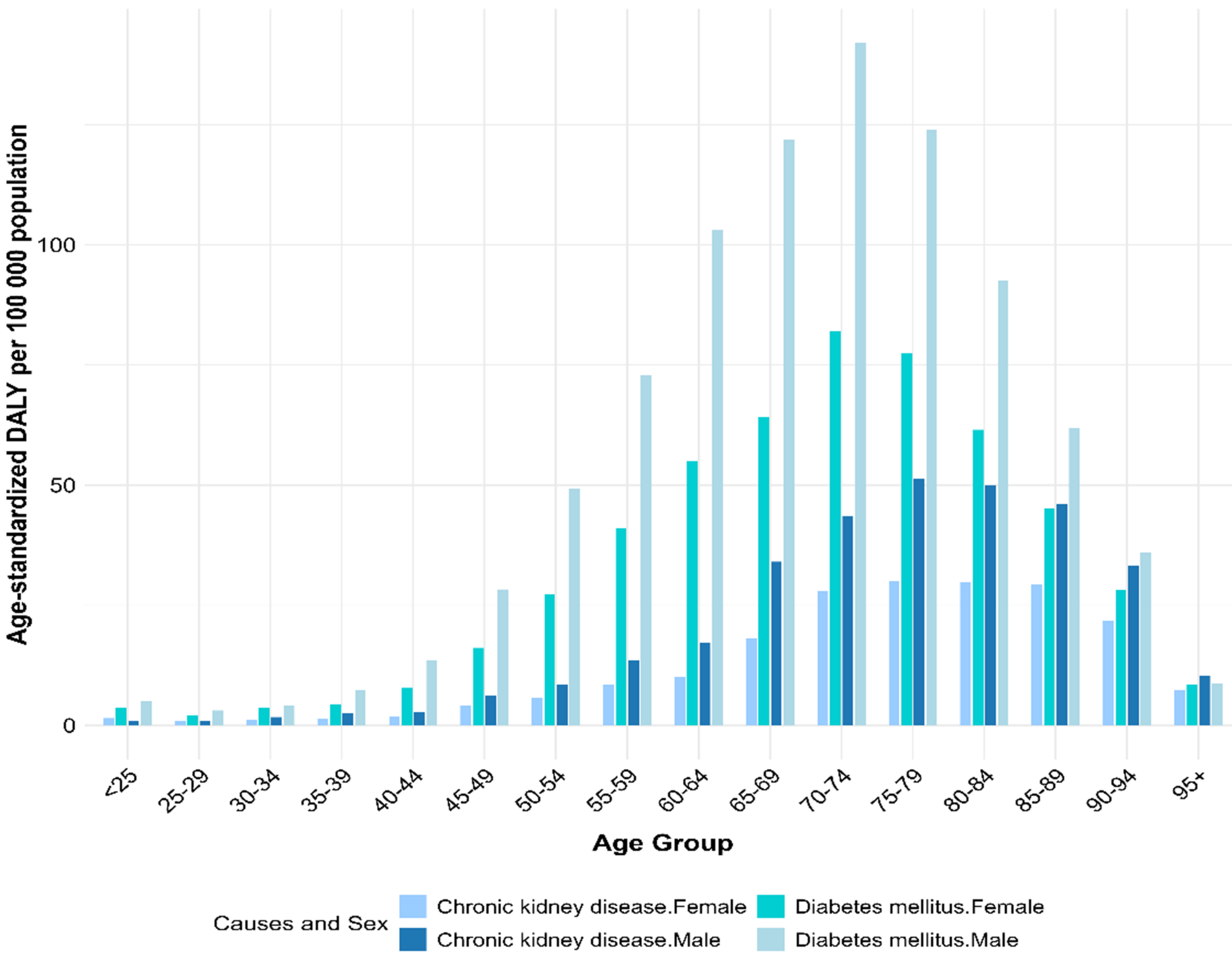

Fig. 1 Age-standardized DALY rates for diabetes mellitus and chronic kidney disease by age group and sex, France 2017

Discussion

Our study provides subnational estimates of DALYs for diabetes mellitus and CKD in France for the year 2017. The results showed that diabetes mellitus had a higher burden than CKD, with sex differences being higher among males than females for both diseases. The burden of both diseases increased progressively with age, reaching a peak among individuals aged 70–74 years. This pattern underscores the strong influence of age on disease burden and highlights important sex-specific differences, with older men showing particularly high rates. The relative contribution of mortality was higher for CKD in both sexes, whereas for diabetes, morbidity contributed more than mortality. To explore regional variation, we analyzed ASDRs for diabetes mellitus and CKD across all French regions. While overall ASDRs were lower in metropolitan France compared to French overseas territories, substantial variation was observed within metropolitan regions. The highest ASDRs were found in the North-East regions, Hauts-de-France and Grand Est, while the lowest were observed in the West and South-West regions, including Bretagne, Pays-de-la-Loire, and Occitanie. Île-de-France, which includes Paris, showed intermediate values relative to other metropolitan regions. In contrast, almost all French overseas territories displayed significantly higher ASDRs for diabetes, Mayotte and Guyane also exhibited the highest ASDRs for CKD. These findings underscore deep-rooted regional health disparities. According to the 2022 report by the Institut National de la Statistique et des Études Économiques (INSEE), extreme poverty is not only more prevalent but also more severe in French overseas territories such as Guadeloupe, Martinique, Guyane, and Mayotte [35].

French overseas territories consistently bear a disproportionate burden of chronic diseases. Structural factors such as entrenched poverty, fragmented health infrastructure, and evolving nutritional challenges likely play a major role [36]. Hypertension, a major risk factor for both diabetes complications and CKD, places stress on the kidneys and accelerates progression to end-stage renal disease [37]. Studies also indicate that sex-specific disparities in disease prevalence and control are more

Fig. 2 Relative contribution of Age-standardized YLL and YLD rates to the total burden of disease per 100 000 (ASDRs) for diabetes mellitus and chronic kidney disease, both sexes, France, 2017

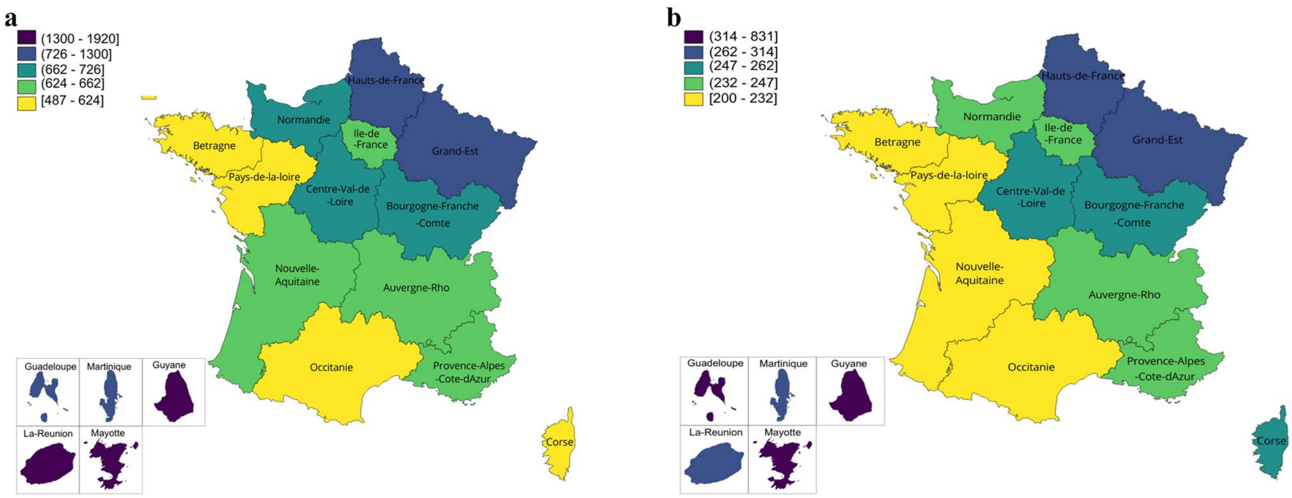

Fig. 3 **a** Age-standardized DALY rates per 100 000 due to diabetes mellitus, both sexes at sub-national regions in France, 2017. **b** Age-standardized DALY rates per 100 000 due to chronic kidney disease, both sexes at sub-national regions in France, 2017

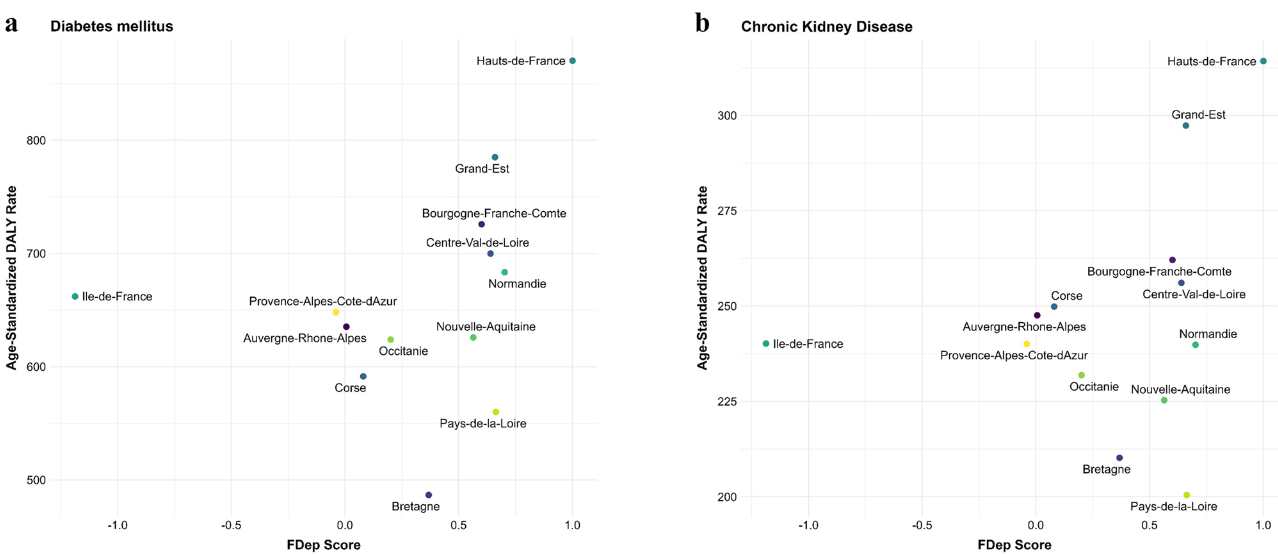

Fig. 4 Correlation between FDep score and ASDRs due to diabetes mellitus (**a**) and chronic kidney disease (**b**) by sub-national regions in metropolitan France in 2017

pronounced in French overseas territories than in mainland France [37, 38]. Socioeconomic vulnerability, such as the high levels of deprivation observed in Guyane, has been linked to poorer metabolic outcomes and may reflect broader regional trends [39]. Futhermore, limited access to health screenings and hypertension management, often leaves elevated blood pressure uncontrolled, aggravating both kidney and diabetes outcomes [37]. High body mass index (BMI) and obesity also substantially increase the risk of diabetes and contribute to kidney disease progression [40]. Rising obesity trends in these territories are strongly influenced by socioeconomic conditions, such as food insecurity and the predominance of inexpensive, calorie-dense, low-nutrient foods [41, 42]. Many communities rely on imported, processed foods high in refined sugars, salt, and unhealthy fats, due to affordability and limited access to fresh produce. Physical inactivity further contributes to these conditions, exacerbated by a lack of safe and accessible spaces for exercise and by cultural or economic barriers to regular physical activity [43]. Socioeconomic disadvantage reinforce these disparities. According to a 2022 report from DREES (Direction de la recherche, des études, de l'évaluation et des statistiques), chronic diseases disproportionately affect individuals with low incomes and significantly reduce life expectancy in France [44]. Geographic isolation, socioeconomic inequalities, and limited healthcare workforce capacity further constrain prevention and management of chronic diseases in the French overseas territories [45–47]. These constraints delay early detection and effective treatment of diabetes and CKD, contributing to persistently poorer outcomes.

A high DALYs value reflects a greater loss of healthy life years due to disease, disability, or premature mortality. In this study, we observed a positive but non-significant correlation between FDep score and ASDRs for diabetes mellitus and CKD across metropolitan France. This result was somewhat unexpected, given the well-established association between socioeconomic deprivation and chronic disease burden. Several factors may explain this finding. *First*, the FDep index aggregates four socioeconomic variables (unemployment rate, blue-collar worker proportion, high school diploma rate, and median income per consumption unit). These components may have compensatory effects at the regional level, where high deprivation in one dimension may be offset by lower deprivation in another, potentially obscuring a consistent relationship with health outcomes. *Second*, the FDep score is constructed usingprincipal component analysis (PCA) to summarize variance in socioeconomic indicators, not to maximize correlation with health outcomes such as DALYs. The resulting score may therefore not align directly with variations in disease burden. *Finally*, important contextual factors that affect health, including healthcare access, environmental exposures, and population density, are not captured in the FDep score but could moderate the relationship between deprivation and disease burden. These results highlight the complexity of interpreting regional health inequalities using composite socioeconomic indices. Given the non-significant ecological correlation between FDep scores and ASDRs, we do not infer an association at the individual level. We acknowledge that ecological analyses are susceptible to ecological fallacy, residual confounding (e.g., age structure, comorbidity burden, healthcare access), and measurement error. Since BoD estimates combine measures of mortality and morbidity, they enable comparisons across conditions, different years and countries. This makes it possible to compare diseases that cause early death with those that lead to long-term illness, while recognizing the limits of the method [48, 49]. These results were comparable with other European countries, conducting national BoD analyses. For instance in Belgium, the ASDR due to diabetes mellitus were lower than in France (526 ASDR vs. 685 ASDR) [50]. In comparison, for CKD, these estimates were slightly higher in Belgium than in France (308 ASDR vs. 252 ASDR) [50]. In Scotland, both ASDR were 744 for diabetes mellitus and for CKD 387 [51]. In Germany, the ASDR for diabetes mellitus was 864 in 2017 [49], however, differences in methodology, such as using national life table for YLL calculation, may partly explain the variation. Compared to these countries, France shows a moderately high burden of diabetes, slightly higher than Belgium but lower than Scotland and Germany. In contrast, the burden of CKD remains comparatively lower. This pattern may reflect differences in risk factor prevalence, healthcare access, or disease coding practices. These international comparisons highlight the importance of standardized methodologies and suggest that targeted regional interventions, particularly for diabetes to reduce disparities, could align with better-performing systems.

Strengths and limitations

This study has some strengths and limitations. This is the first study considered as an initial proof-of-concept, to estimate the DALYs due to diabetes mellitus and CKD at the regional level for the whole France including all metropolitan regions and French overseas territories. We used all available relevant national data sources to capture the burden of these diseases, which have the best representation of the whole French population [20]. We also compared the SNDS-based prevalence estimates with a national cohort CONSTANCE to validate the coverage. Our results align closely and strenghthen the validity of our prevalence estimates. More detail is provided in the *Supplementary material 1 : Additional file 4.* However, some limitations should be acknowledged. *First*, our estimates for diabetes prevalence were based on pharmacologically treated cases in the SNDS, which was designed for administrative rather than epidemiological purposes. While this approach excludes untreated (~ 1.2%) and undiagnosed (~ 1.7%) diabetes, these proportions are relatively small and likely have only a modest impact on our results [52]. *Second*, we used disability weights and severity distributions derived from the Global Burden of Disease (GBD) 2019 study. While these may not fully capture the French population's specific disease severity perceptions or healthcare context, they provide a standardized and internationally validated framework that ensures comparability across countries and conditions. In the absence of France-specific disability weights and severity distributions, applying GBD estimates represents the most robust and widely accepted approach for national burden of disease assessments. Nevertheless, this limitation should take into account while interpreting our results. *Third*, we did not apply a correction for multimorbidity between diabetes mellitus and CKD, which may have led to overestimate of DALYs in individuals affected by both conditions [53]. This limitation should be considered when interpreting the results. Future research should incorporate methods to adjust for comorbidities to refine the disease burden estimates. *Fourth*, individuals who were not insured (e.g., homeless and illegal people, foreigners with their official residence abroad), were not included in the database. It may have an impact on the YLD, hence DALY estimates. *Finally*, this analysis focused solely on regional-level variation. The regions are generally highly heterogeneous, including both rural and urban areas, as well as significant variation in access to healthcare within the same region. This heterogeneity may also explain the lack of correlation with the FDep; for instance, in the case of the Ild-de-France region. Future research should incorporate factors such as rurality, population density, and healthcare access to allow for more granular geographic stratification and deeper insight into spatial health disparities. These limitations may have led to an under- or overestimation of the disease burden in certain regions, and they highlight the need for further research using refined methodologies.

We identified the following data gaps *(implications for method)* to improve these data sources (i.e., SNDS and REIN registry) for future estimates: *First*, it is recommended to link the data on medical diagnoses during outpatient consultations or visits or national surveys or cohort studies, with the SNDS that may help to identify the severity levels of various diseases at the regional level. *Second*, the disability weights for diabetes with gestational diabetes and prediabetes are required to be calculated. *Third*, the REDSIAM algorithm will be improved by adding a new approach based on artificial intelligence algorithms and by using various databases, which would lead to a better define the different stages of CKD. *Fourth*, the disability weight for end-stage kidney disease (ESKD), with kidney transplant is very low (0.024 [0.014–0.039]), whereas in reality, these patients suffer more and have more difficulties with daily activities. Therefore, it is recommended to update it. *Finally*, for severity proportions of diabetes mellitus, we extracted data for France from the GBD 2019 study and calculated the relevant proportions of cases by severity level. The main limitation of using these proportions is that are mostly remain constant over time [54]. Nevertheless, the use of national results is preferable as previous studies have shown striking differences between globally constant versus national distributions [55].

Implications in public health

These findings underscore the need for targeted and region-specific prevention strategies. Structured diet and physical activity programs, such as community-based group sessions led by trained personnel, have proven feasible and cost-effective in reducing type 2 diabetes risk [43, 56, 57]. Adapting such interventions to the specific needs and contexts of the French overseas territories is essential. In these territories, service delivery challenges remain a major concern. In Guyane, for example, general practitioners report barriers to prescribing physical activity due to limited training and inadequate infrastructure [43]. Addressing these challenges requires sustained local investment and tailored public health actions. Priorities include strengthening health systems, improving healthcare accessibility and professional training, and promoting healthier food environments through community-based prevention initiatives. Although based on 2017 data, the burden estimates from this study remain highly relevant for the current public health planning. They provide a robust pre-pandemic baseline for understanding the regional burden of diabetes mellitus and CKD in France. Despite shifts in public health priorities during the COVID-19 pandemic, these chronic diseases continue to represent a substantial and increasing share of long-term morbidity and healthcare costs.

These results have important implications for both policy and future research. *Policy implications* : Given the high burden of diabetes mellitus and CKD on population health, integrated and collaborative approaches are needed [58]. Strengthening coordination between healthcare providers, particularly endocrinologists and nephrologists, can optimize disease management, protect kidney function, and reduce diabetes-related complications. *Future research* should incorporate methods to adjust for comorbidities, in order to refine the disease burden estimates and further examine the contribution of major risk factors to disability. Socioeconomic determinants should be assessed using more granular indicators, such as individual-level socioeconomic status measures (e.g., income, education, and occupation) and healthcare accessibility. In addition, investigating potential nonlinear relationships and multilevel influences may help address the limitations of ecological measures like the FDep. Forecasting future regional trends in disease burden would also be valuable for public health planning. Updating these estimates with post-pandemic data could reveal whether disruptions to routine care, uneven telemedicine adoption, differential broadband access, and regionally heterogeneous economic shocks have altered chronic disease management and regional disparities.

Conclusions

These results highlight the importance of population health surveillance of diabetes mellitus and CKD in France according to age, sex and region in 2017, by taking into account the claim data and kidney registry data. These results could be useful to inform decision-making processes in health policy such as recommendations to implement appropriate health interventions and evidence-based age- and sex-specific prevention strategies at regional levels, which can have major health benefits. The framework adopted to calculate DALYs for selected diseases can be extended to additional diseases. This approach holds the potential for significant and lasting benefits by reducing the prevalence and impact of these major diseases across diverse populations in France. Future research should take into account the contribution of various risk factors and investigate the socioeconomic determinants of health to better understand the mechanisms underlying health disparities in these diseases at subnational level.

Abbreviations

BoD	Burden of Disease
YLL	Years of Life Lost
SEYLL	Standard Expected Years of Life Lost due to premature mortality
YLD	Years lived with disability
DALY	Disability adjusted life years
GBD	Global Burden of Disease
SNDS	Système National des Données de Santé
REIN	Le Réseau Epidémiologique et Information en Néphrologie
CKD	Chronic kidney disease
RTT	Renal transplant therapy
ESRD	End stage renal diagnosis
RENALGO	RENal ALGOrithm, REDSIAM: Réseau pour mieux utiliser les Données du Système national des données de santé
FDep	French Deprivation Index
PCA	Principal Component Analysis
UI	Uncertainty interval
INSERM	Institut National de la Santé et de la Recherche Médicale
CépiDC	Centre d'épidémiologie sur les causes médicales de Décès
INSEE	Institute National de la Statistique et des études économiques
ASPR	Age-standardised prevalence rate
ASDR	Age-standardized DALY rate
BMI	Body Mass Index

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13690-025-01767-1.

Supplementary Material 1. Additional file 1: It describes the case definition and details of the algorithms used to identify individuals with diabetes mellitus and CKD in SNDS. Additional file 2: It describes the GBD disability weights for diabetes mellitus and chronic kidney disease. Additional file 3: It describes the severity proportions diabetes mellitus and chronic kidney disease. Additional file 4: It describes the comparison of the SNDS-based prevalence estimates with a national cohort CONSTANCE-bases prevalence estimates.

Acknowledgements

The authors would like to acknowledge the technical networking support from COST Action CA18218 (European Burden of Disease Network: www.burden-eu.net: Under European Cooperation in Science and Technology). We gratefully acknowledge all REIN registry participants. The centers participating in the registry are listed in the REIN annual report: https://www.agence-biomedecine.fr/Les-chiffres-du-R-E-I-N.

Authors' contributions

CC and RH generated the idea for the study. CC, MR and RH developed the methodological approach. CC and MR supported data curation. CC and RH carried out all the analyses. NM and RH contributed to the visualization of results. CC and RH drafted the original manuscript. NM, E.v.d.Lippe, BD, and G.M.A. Wyper provided critical input into the methodology and interpretation of the results. All other authors reviewed and approved the final draft of the manuscript.

Funding

This research received no specific grant from any funding agency in the public, profit or non-profit sectors. NM was supported by a grant funded by the National Research, Development and Innovation Fund of Hungary (research project 143383).

Data availability

The datasets generated and/or analysed to construct the figures in the current study, are available in the manuscript and supplementary material 1.

Declarations

Ethics approval and consent to participate

A specific ethics committee approval was not required for this study. The French REIN registry and the Institute of Biomédecine have permanent access to the pseudonymized reimbursement data SNDS (système national des données de santé) in application of the provisions of articles R. 1461–12 et seq. of the French Public Health Code with rules and this study adhered to the Declaration of Helsinki. The study was based on aggregated and anonymized data and did not require the informed consent to participate.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 10 July 2025 / Accepted: 29 October 2025
Published online: 10 December 2025

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